
Esimbi

Here are two sequences of five numerals written in Esimbi¹. In each sequence, the ratio between each number and the next is the same. Note that this ratio may or may not be different across the two sequences.

Sequence 1

- (a) mərakpə
- (b) moŋi
- (c) moŋioŋi
- (d) buyu nə malalə
- (e) mobuyu makələ nə mərakpə

Sequence 2

- (a) kenə
- (b) makələ
- (c) matənə moŋi
- (d) mobuyu mərakpə nə matənə mərakpə
- (e) mobuyu moŋioŋi nə kenə

Assignment 1:

Find the Hindu-Arabic numerals (i.e. the numbers you usually write) that correspond to the two sequences above.

Assignment 2:

Following are three expressions in Esimbi, where the blank boxes are placeholders for the missing operators. All three expressions evaluate to the same number, 42.

(a) mobuyu buyu nə mobuyu mərakpə nə malalə makələ

(b) mobuyu matənə nə kenə matənə moŋi

(c) mobuyu mərakpə nə kenə mərakpə

Write 42 in Esimbi numerals. Fill in the missing operators and write the equations in Hindu-Arabic numerals.

– Mukund Choudhary and Aparna Gupte

Note: Since PLO 2021 Round 1 was held as an online contest, this problem was presented in an obfuscated form, with the real facts about the language modified/omitted.

¹ Esimbi is a Tivoid language of southwestern Cameroon. It is also called Isimbi, Simpi, Age, Aage, Bogue and Mburugam.